



NITI Aayog



SECTOR AT A GLANCE 07

Emerging Technologies in India

Energy in India

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EMERGING TECHNOLOGIES IN INDIA

ENERGY IN INDIA

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Emerging technologies in the energy sector are opening new possibilities for sustainable and clean energy sources. In India, these span synthetic hydrocarbon fuels, multiple new battery technologies (including lithium-air and potassium-ion batteries), carbon capture, utilization, and storage technologies, coal gasification, and small modular reactors. The Ministry of New and Renewable Energy (MNRE) focuses its programs on four new technologies: ocean energy, geothermal energy, hydrogen energy, and energy storage technologies like batteries, fly wheels, pump hydro, and capacitors to develop India's indigenous research and industrial base. This document focuses on ocean energy and geothermal energy, while hydrogen energy, negative emission technologies, and energy storage are discussed in separate Sector-at-a-Glance documents.¹ Both ocean and geothermal energy are in relatively nascent stages, largely focusing on exploration. Geothermal energy provides a stable and continuous source of clean energy by harnessing the heat generated from the Earth's core, while ocean energy utilizes the power of waves, tides, and temperature differentials to generate electricity.

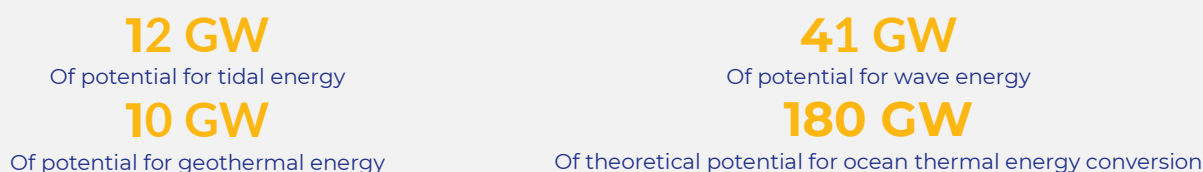
Geothermal energy stands out as a reliable and consistent source of power for the future, unaffected by weather conditions unlike solar and wind energy. This inherent stability ensures a continuous and uninterrupted supply of energy. Ocean energy is quite predictable owing to the cyclic patterns of waves, tides, and currents, which enables it to complement other renewable energy sources. With policy and investment support, emerging technologies can drive substantial power generation over the coming decades.

Nevertheless, challenges such as high generation costs, the complexity of assessing the potential of ocean energy, a shortage of skilled workforce, and insufficient demonstration projects impede private sector involvement and hinder the widespread adoption of these technologies at a commercially viable scale.

Given these opportunities and challenges, fostering indigenous R&D and demonstration of these technologies is the major policy focus area. Several ongoing and recent central government initiatives, including but not limited to the Mission on Advance and High-Impact Research focus on conducting research and demonstration of emerging technologies in the power sector. These efforts, along with several state- and local-level policies and initiatives, seek to harness emerging technologies to realize India's energy transition goals.

Promoting R&D through global and domestic collaborations, developing an enabling policy environment for greater deployment, establishing key physical and market infrastructure, and advancing private participation are efforts critical to continue advancing emerging energy technologies. This note presents key trends related to different components of ocean and geothermal energy in India and actions for the way forward.

Figure 1: Key emerging technologies statistics

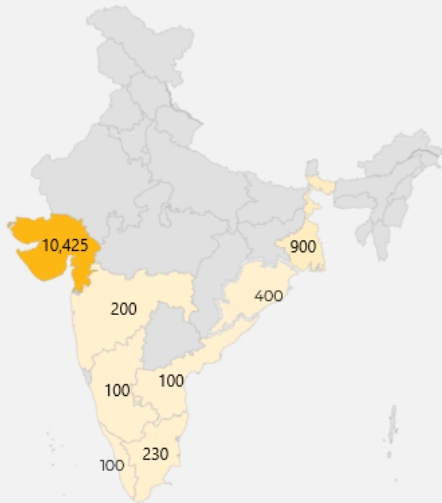


Source: Parliamentary Standing Committee on Energy (2020-21, Tidal Power Development in India, 2021; PIB, Power Minister Shri RK Singh approves proposal to declare ocean energy as Renewable Energy, 2019

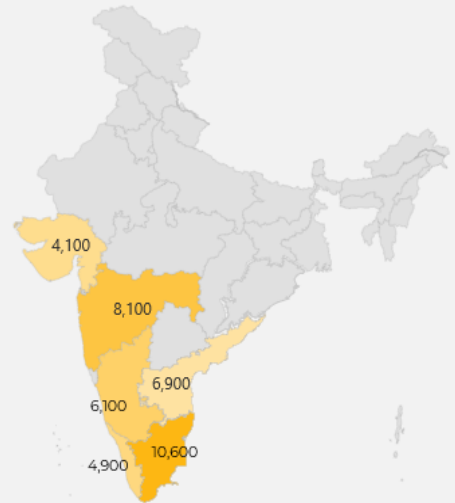
¹ Ministry of New and Renewable Energy, New technologies, Accessed August 2023; for a discussion of emerging technologies and fuels, see IEA, ETP Clean energy technology guide, 2023; for a discussion of future energy efficiency technologies, see BEE, Unlocking National Energy Efficiency Potential – UNNATEE, Strategy plan towards developing an energy efficient nation (2017-2031), 2018

Figure 2: Estimated ocean energy potential in India

Estimated theoretical potential of tidal power (MW)

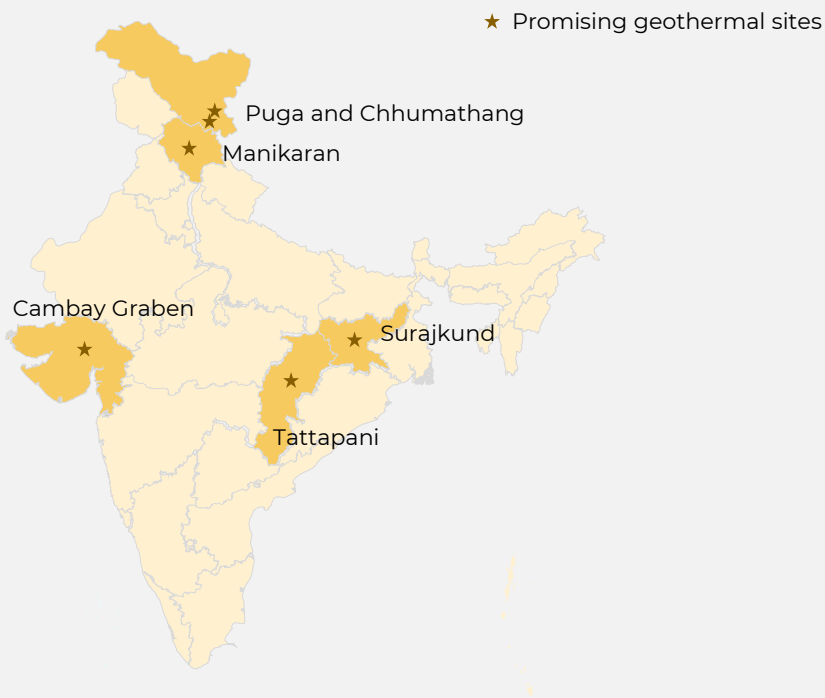


Estimated theoretical potential of wave power (MW)



Source: Standing Committee on Energy: Seventeenth Lok Sabha, Ministry of New and Renewable Energy – Tidal Power Development in India: Twentieth Report, 2021
 Note: The last primary component of ocean energy is ocean-thermal energy. Its potential has not been estimated by the Ministry of New and Renewable Energy.

Figure 3: Most promising geothermal sites for geothermal energy development in India



Source: MNRE, Draft National Policy on Geo-Thermal Energy, accessed December 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by increasing the adoption of ocean and geothermal energy. This primarily includes the contribution of these technologies in advancing India's transition to clean energy.



India's geographic resources enable significant potential for emerging technologies:

India's vast coastline spanning over 7,500 km presents a significant opportunity for harnessing ocean energy resources. Promising areas for tidal and wave energy technologies include the western coast of Maharashtra, Goa, Karnataka, and Kerala among others. According to a study by IIT Madras, India has a theoretical potential of 12 GW for tidal energy and 41 GW for wave energy. Similarly, for geothermal energy, the Geological Survey of India (GSI) has identified approximately 300 geothermal hot springs across the country. Jammu & Kashmir, Gujarat, Maharashtra, and other regions show promise for geothermal power generation.² The geothermal hot springs across the country exhibit a potential of 10 GW. Most of these hot springs fall into the medium potential (100°C to 200°C) and low potential (<100°C) categories. In 2019, it was estimated that these three energy sources collectively had the potential equivalent to 15% of India's installed power generation capacity.³



Achieving clean energy transition goals, energy efficiency and security:

Emerging technologies like geothermal energy and ocean energy have the potential to contribute towards advancing India's transition to cleaner and renewable energy sources and reducing India's dependence on fossil fuels, and promoting energy efficiency. To enhance ocean energy utilization in India, the generation of energy through diverse forms of ocean resources like tidal, wave, ocean thermal energy conversion, etc., can qualify to fulfil non-solar Renewable Purchase Obligations (RPO).⁴ Further, pilot efforts are already underway to use geothermal energy for power generation.⁵ By diversifying the energy mix with these indigenous and clean sources, India can reduce its dependence on imported fossil fuels, leading to greater energy security.

Challenges

This section highlights some of the major challenges and constraints facing ocean and geothermal energy in India.



Energy assessment challenges:

According to a report from IIT Bombay, there are substantial difficulties in evaluating the potential of ocean energy in India. These challenges arise from the extensive coastal length, diverse dynamics, geography, seasonal and regional wave patterns and characteristics, limitations in on-site measurements, and complexities associated with assessing environmental impacts.⁶

² Ministry of New and Renewable Energy, Geo-Thermal and Ocean Energy Technologies, 2019

³ Ministry of New and Renewable Energy, Geo-Thermal and Ocean Energy Technologies, 2019

⁴ Ministry of New and Renewable Energy, Power Minister Shri RK Singh approves proposal to declare ocean energy as Renewable Energy, 2019

⁵ PIB, Diversification of Coal CPSE's for Sustainable Energy Ecosystem, 2024

⁶ Indian Institute of Technology, Bombay, Estimating Ocean Renewable Energies for Potential Harvesting, Accessed July 2023



High generation cost and infrastructural constraints:

The current cost of energy generation from emerging technologies is prohibitively high relative to other mature technologies or sources. Harnessing energy from the ocean and geothermal resources requires specialised technologies and infrastructure, which can be expensive to develop and maintain. Previous efforts to harness ocean energy were unsuccessful due to high capital costs ranging INR 30-60 crore per MW.⁷ Other infrastructural issues such as power evacuation using submersible electricity carrying cables, distance of the plant from the shore, provisioning of offshore platform for plant installation also increases the overall cost of electricity generation. Similarly, drilling and extracting heat from deep geothermal wells can be costly, with an upfront cost of Rs 30 crore/MW. Given their current maturity levels, these sources are not cost-competitive.⁸



Limited workforce capabilities:

Apart from significant investments, the development and implementation of the ocean and geothermal energy projects require specialised knowledge, skills, and technologies that are still nascent in the country. The scarcity of trained professionals and technicians with expertise in ocean energy systems, such as wave and tidal devices, as well as geothermal technologies, hinders the progress in harnessing these resources to their full potential.⁹



Need for private sector engagement and intersectoral coordination:

There are concerns around the development of these technologies due to associated financial, technological, environmental, and other risks. Previous endeavours to establish tidal power projects in India have fallen short primarily due to techno-economic considerations. Additionally, navigating complex intersectoral coordination challenges can be a significant hurdle. The involvement of multiple ministries and agencies, each with its approval processes, creates delays and uncertainties that discourage private investment.¹⁰

⁷ Ministry of New and Renewable Energy, Ocean Energy, Accessed July 2023

⁸ Ministry of New and Renewable Energy, Geo-Thermal and Ocean Energy Technologies, 2019

⁹ Ministry of New and Renewable Energy, Geo-Thermal and Ocean Energy Technologies, 2019

¹⁰ Mongabay, India's tidal power potential hampered by high costs and environmental risks, 2021

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee emerging technologies in the country.



NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. It guides policy development by analyzing the potential of various technologies like geothermal, ocean, and advanced energy storage solutions. NITI Aayog also plays a key role in fostering collaboration between government, research institutions, and the private sector to accelerate R&D and pilot projects that can pave the way for the commercialization of these promising technologies.



Ministry of New and Renewable Energy (MNRE)

The ministry develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security.



Ministry of Earth Science

The Ministry of Earth Sciences in India, overseeing oceanographic research, plays a vital role in the exploration and harnessing of ocean energy resources. The ministry contributes to advancing sustainable energy solutions from the vast potential of the ocean aimed at leveraging this renewable resource for a cleaner and more sustainable energy future.



Ministry of Power

The ministry administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs.



Department of Atomic Energy (DAE)

DAE is responsible for the entire spectrum of activities related to nuclear science and technology. This encompasses construction and operation of nuclear reactors, research and development of nuclear technologies, and exploration of atomic minerals.



Ministry of Coal

The Ministry plays a vital role in supporting research into clean coal technologies like carbon capture, utilization, and storage (CCUS) to minimize the environmental impact of coal-based power generation.

Major policies and programs

This section highlights existing policies related to integrated energy assessments and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Mission on Advanced and High-Impact Research (MAHIR):

The mission, launched jointly by the Ministry of Power and the Ministry of New and Renewable Energy, strives to foster indigenous research, development, and demonstration of the latest and emerging technologies in the power sector. The mission aims to harness emerging technologies and advance them to the

implementation stage, with the goal of utilizing them as the primary catalyst for India's future economic growth and establishing the country as a global manufacturing hub. The following technologies are prioritized for research: alternatives to lithium-ion storage batteries, modifying electric cookers and pans to suit Indian cooking methods, green hydrogen for mobility, carbon capture, geothermal energy, solid-state refrigeration, nano-technology for EV batteries, and indigenous cold-rolled grain-oriented steel technologies.¹¹



R&D support mechanisms:

The Ministry of New and Renewable Energy (MNRE) has initiated research, development and demonstration projects at various research, scientific and educational institutes, universities, etc., to develop indigenous research and industrial base, expertise, and prototype devices for new technologies.¹² The Ministry of Earth Sciences has established the National Institute of Ocean Technology (NIOT) to focus on research and development in ocean-related technologies, including wave energy, hydrokinetic energy and ocean thermal energy conversion (OTEC).¹³

¹¹ Ministry of Power, Mission on Advanced and High-Impact Research (MAHIR) launched to leverage Emerging Technologies in Power Sector, 2023

¹² Ministry of New and Renewable Energy, New Technologies, accessed July 2023

¹³ Ministry of Earth Sciences, National Institute of Ocean Technology (NIOT), accessed July 2023

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the NITI Aayog and the Ministry of New and Renewable Energy.

 Promote R&D through global and domestic collaborations	- Foster partnerships with premier domestic research institutions to establish dedicated R&D centres for ocean and geothermal energy. These centres should focus on technology innovation, feasibility studies, and pilot projects. - Encourage private companies, especially those in the renewable energy domain, to invest in R&D initiatives. Leverage existing government schemes like the production-linked incentive scheme to attract private investment in clean energy technologies. - Facilitate knowledge exchange by organizing workshops, conferences, and technology transfer programs. Collaborate with international energy research and capacity-building bodies to enhance technical expertise and build a skilled workforce.
 Promote enabling policy for deployment and scaling	- Simplify the regulatory process for ocean and geothermal projects, such as single-window clearance mechanisms to expedite approvals for project developers. Adopt successful global best practices, such as Iceland's streamlined geothermal permitting and regulatory process. - Develop a comprehensive national strategy that outlines specific targets for ocean and geothermal energy deployment. Align these targets with India's commitment to the Paris Agreement and the Sustainable Development Goals (SDGs). Regularly review progress and adjust policies accordingly. - Consider the introduction of financial incentives such as tax breaks, feed-in tariffs, and subsidies for ocean and geothermal projects.
 Establish enabling physical and market infrastructure	- Invest in accurate resource assessment and mapping for ocean and geothermal potential across India. Collaborate with local and international think tanks and research institutes to create detailed resource databases. - Prioritize infrastructure development, including research vessels for ocean energy assessment and drilling rigs for geothermal exploration. Explore public-private partnerships to fund critical infrastructure projects. - Establish risk-sharing mechanisms such as insurance schemes, public guarantees, and innovative financing models to attract investors and reduce perceived risks.
 Enhance private sector engagement	- Engage with industry associations to create awareness about ocean and geothermal technologies. Encourage joint ventures and technology transfer agreements. - Create a conducive market environment by promoting ocean and geothermal energy through pilot projects, demonstration plants, and public awareness campaigns. Leverage the Indian Energy Exchange (IEX) and power trading platforms to integrate these technologies into the grid. - Collaborate with educational institutions and vocational training centres to develop specialized courses on ocean and geothermal energy, and

	encourage skill development and certification programs to address the workforce shortage for such technologies.
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Recommended readings

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